企業電腦網路



Chapter 1

Welcome to the Cloud

中央資管

陳奕明

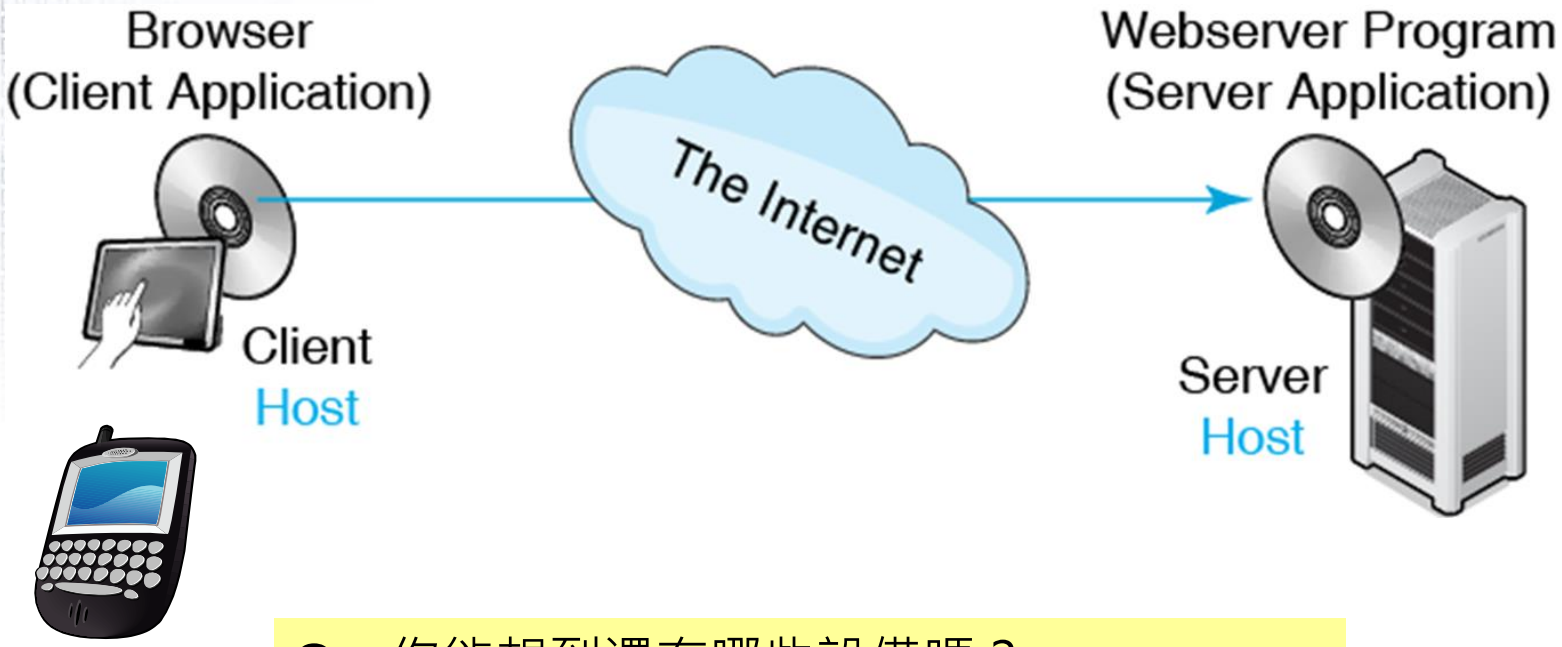
Why Networking is a Head Game?

- There is a lot of information to master
- There are many TLAs (three-letter acronyms), e.g. OSI, PPP, TCP, SLA *et al*
- Why? Because when designing and troubleshooting, you must know everything of anything.
  - One piece of ignorance can ruin the whole project

- This chapter is a survey of the key concepts we will see in this course
- Netflix Jumps Into the Amazon
- Messages
- Single Networks
- Internets
- LANs vs. WANs
- Standards Layers

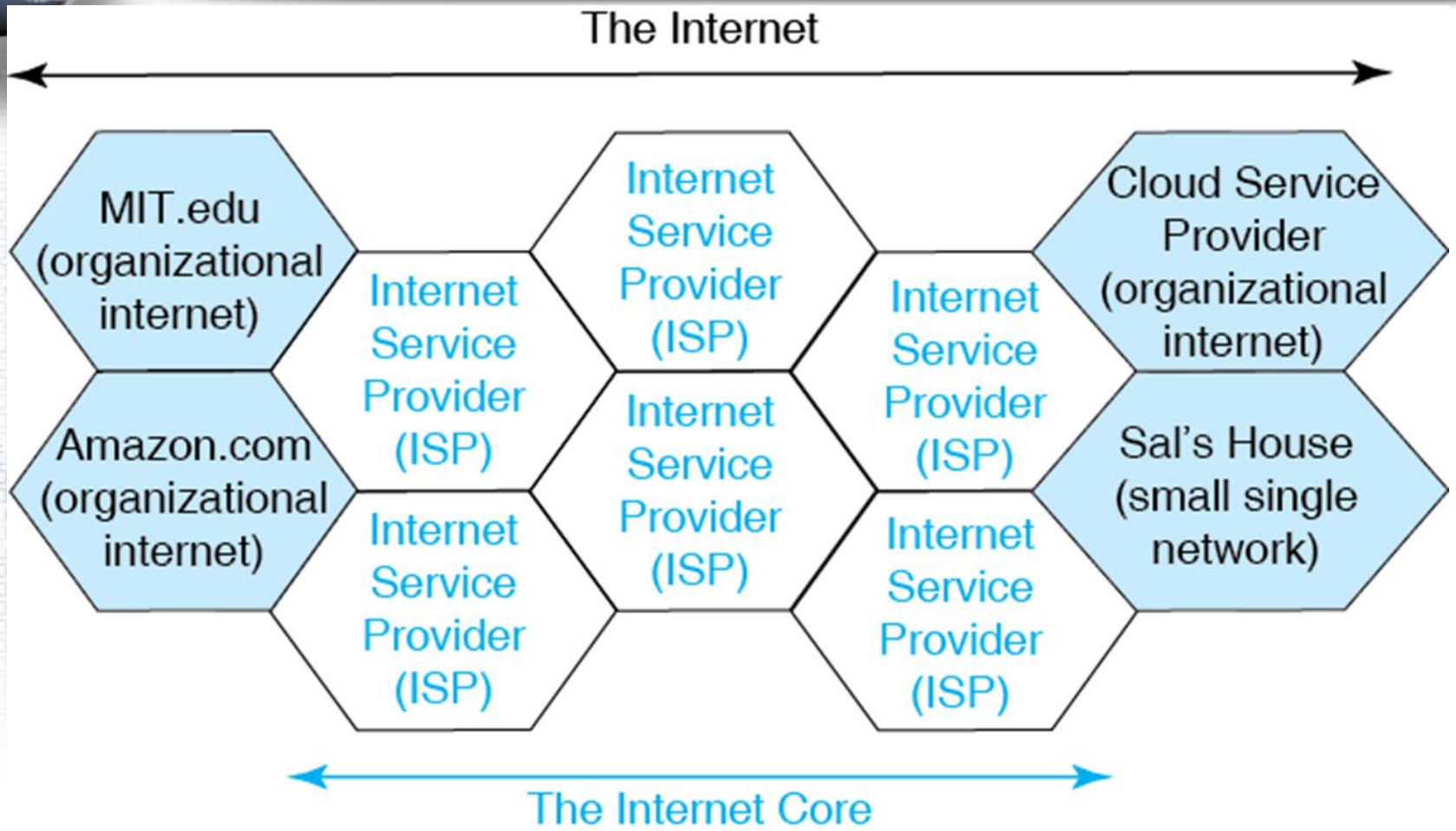
Internet communications

- What Is a Network?
  - A communication system that allows application programs on different hosts to work together (Hosts can be client PCs, servers, mobile phones, etc. [註])



Q：你能想到還有哪些設備嗎？

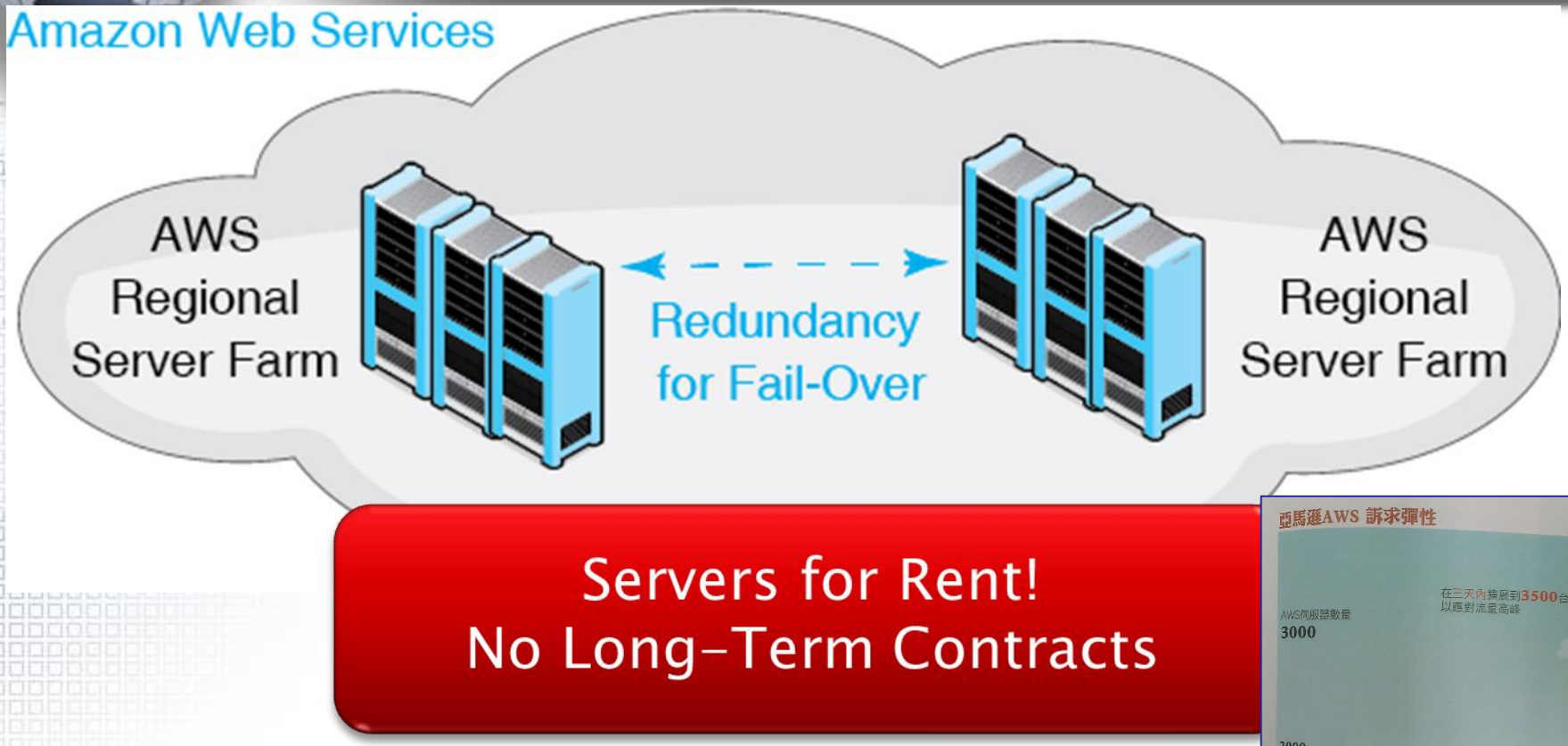
1.3 The Internet

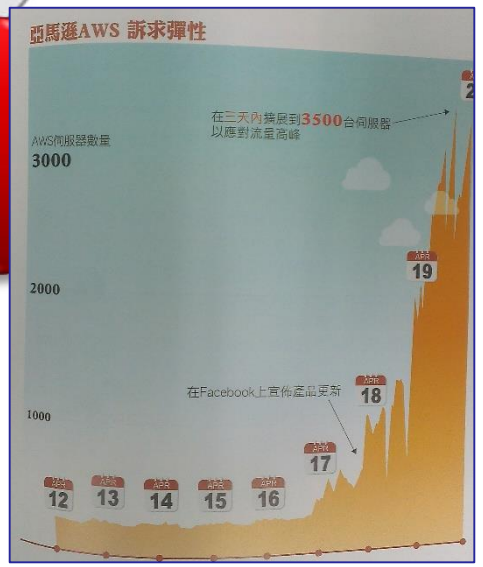


Q:你知道上圖中間的ISP是哪些公司嗎？
中華電信HiNet位置在哪裡？
Q：你聽過「第一類電信業者」「第二類電信業者」？你知道兩者區別嗎？

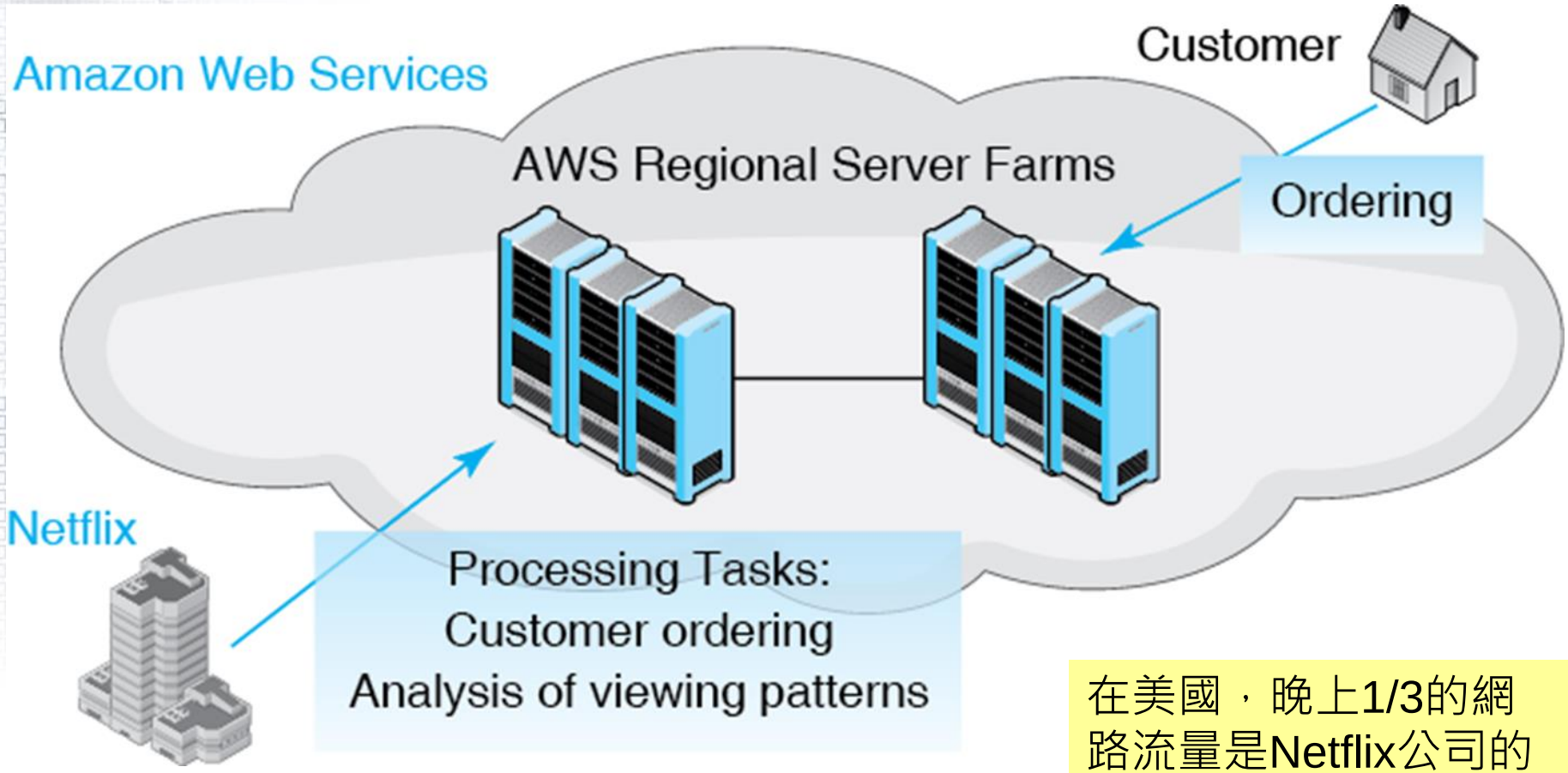
Q:當你在學校的教室上網，你知道你連接的ISP是誰嗎？

1.4 Netflix and Amazon Web Services -Amazon side



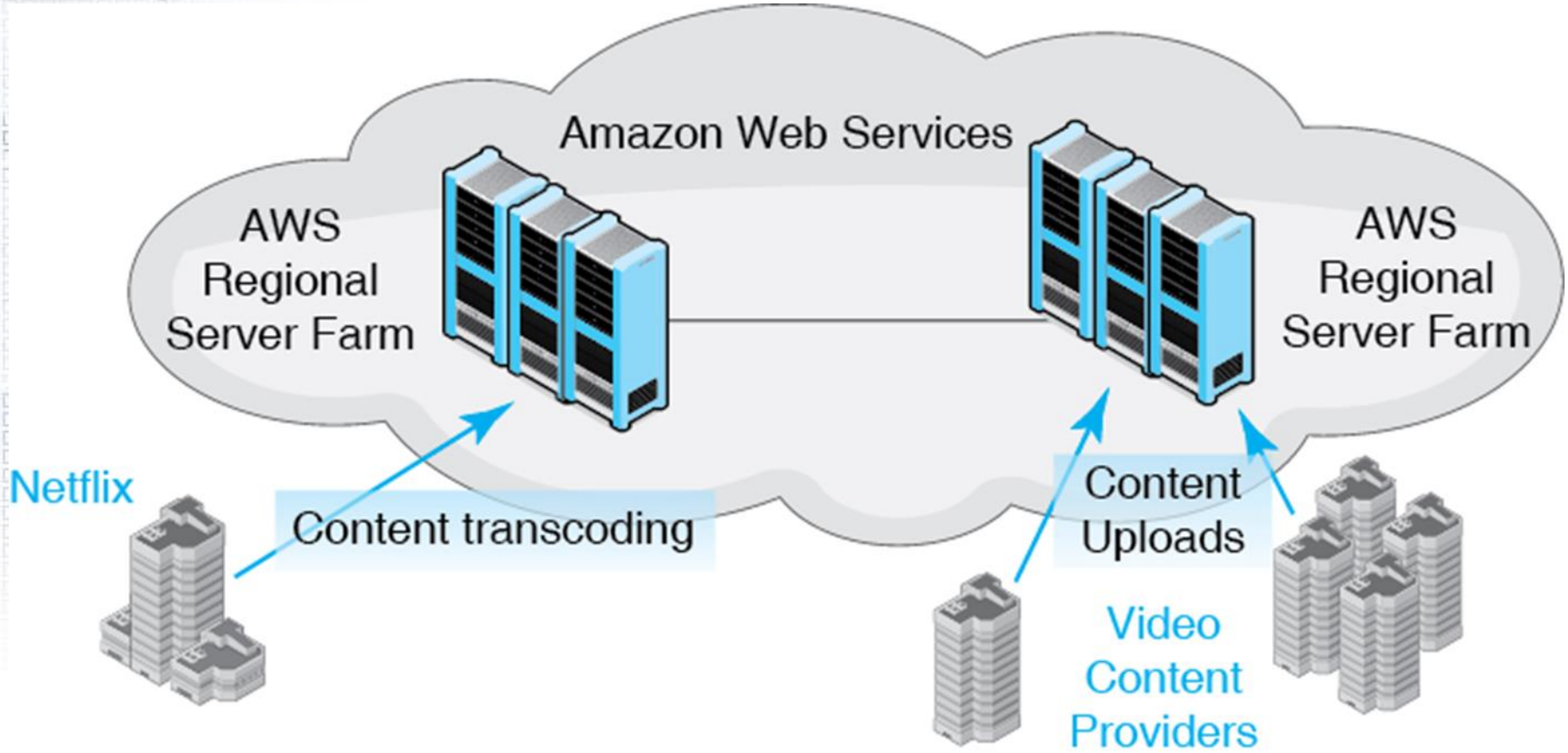


1.4 Netflix and Amazon Web Service - Netflix side

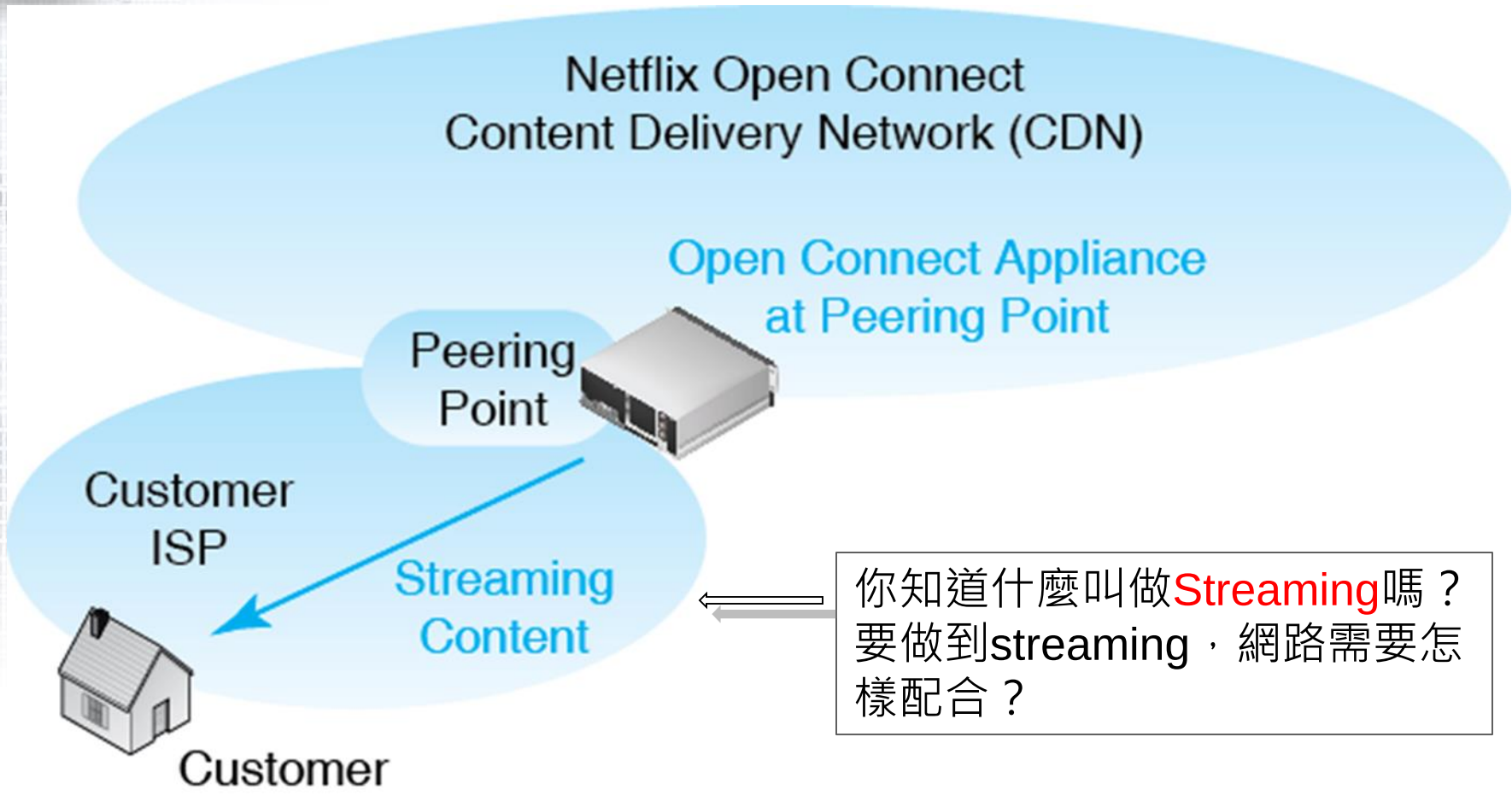


在美國，晚上1/3的網路流量是Netflix公司的

1.4 Netflix and Amazon Web Services -3rd party: Content Provider

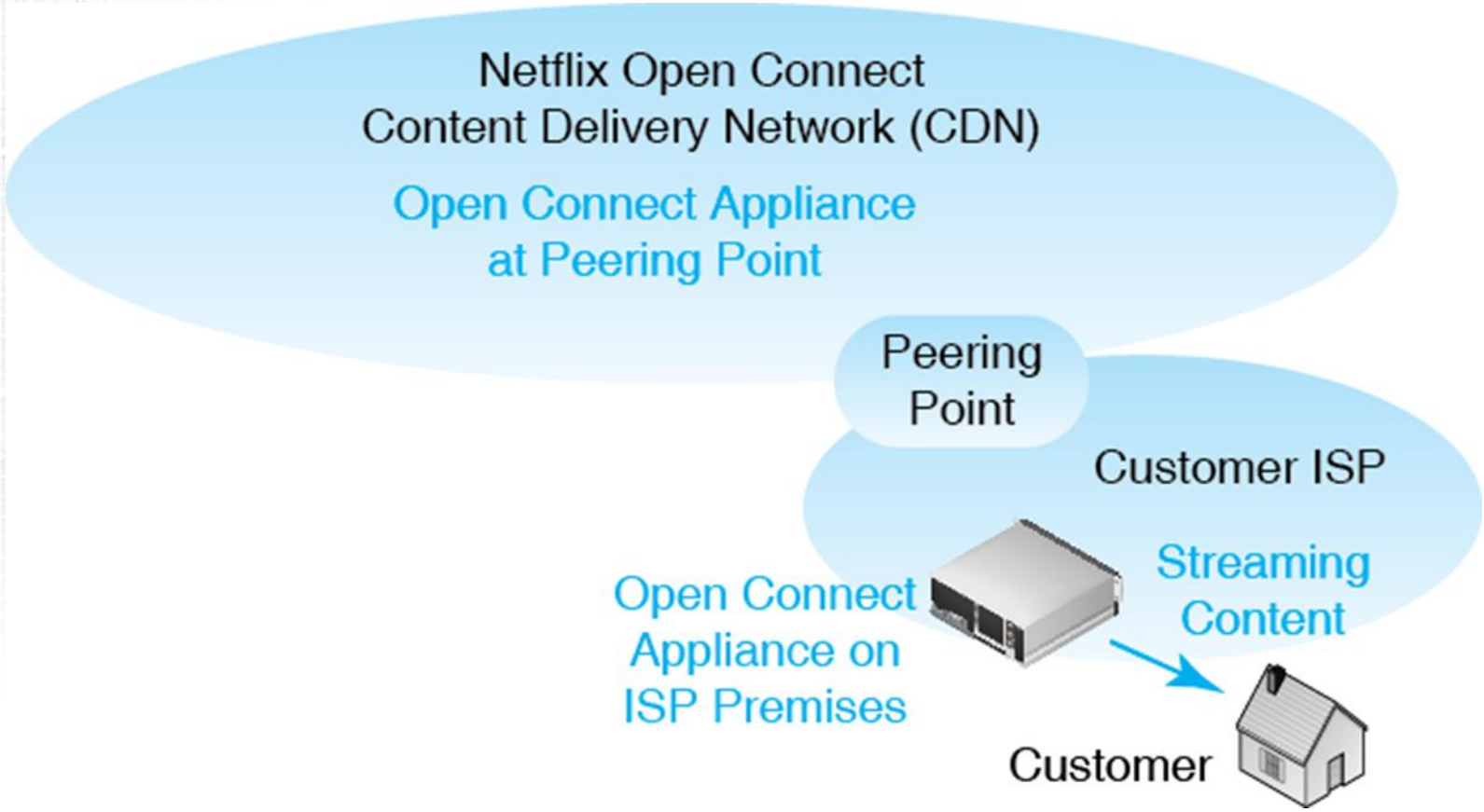


1.5 Netflix Open Distribution Network -Functions of CDN and P2P (1/2)



你知道什麼叫做Streaming嗎？要做到streaming，網路需要怎樣配合？

1.5 Netflix Open Distribution Network -Functions of CDN and P2P (2/2)



Inside Amazon
-Rack Servers and Equipment Rack









Equipment Rack

Rack Servers

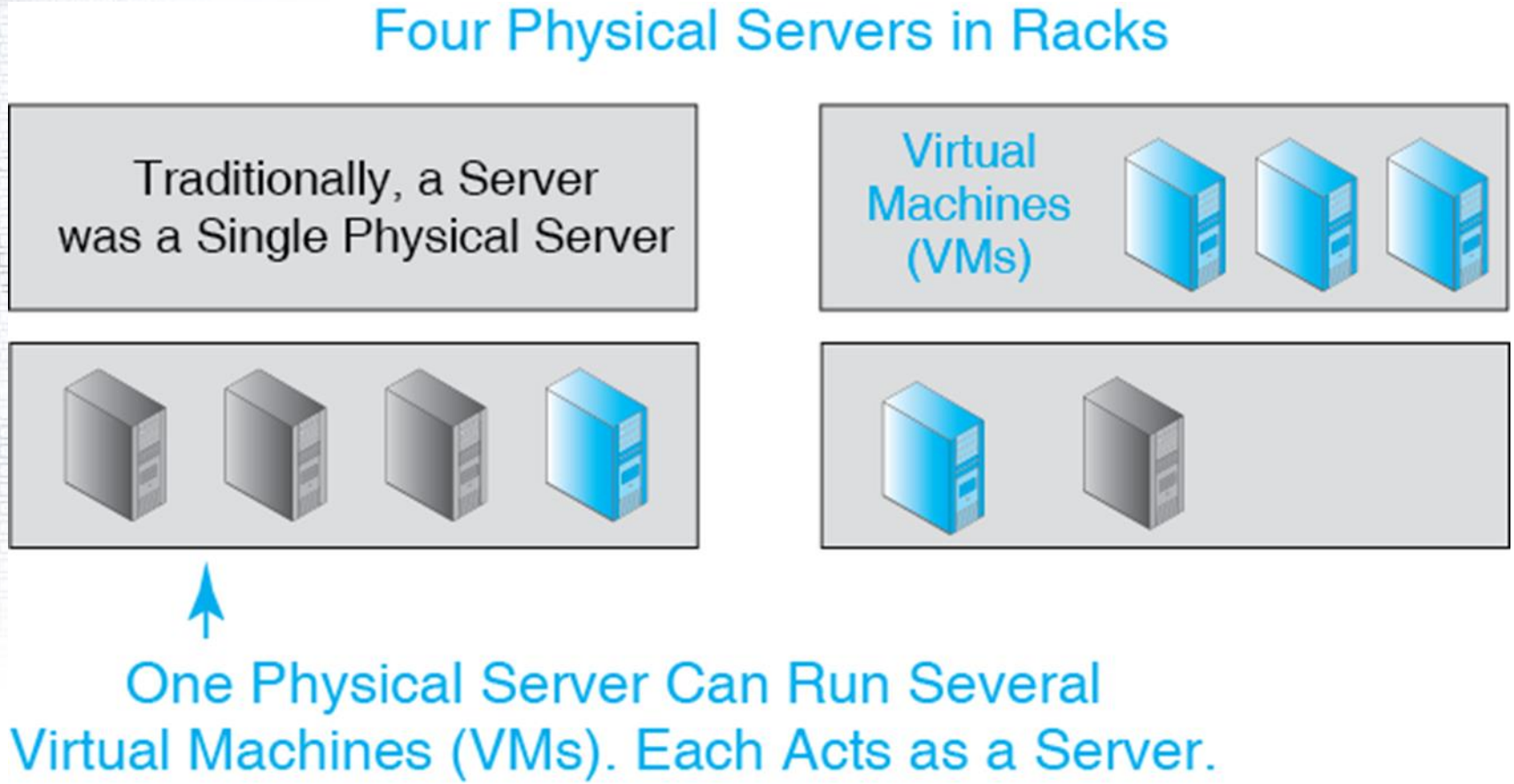
Individual Rack Server



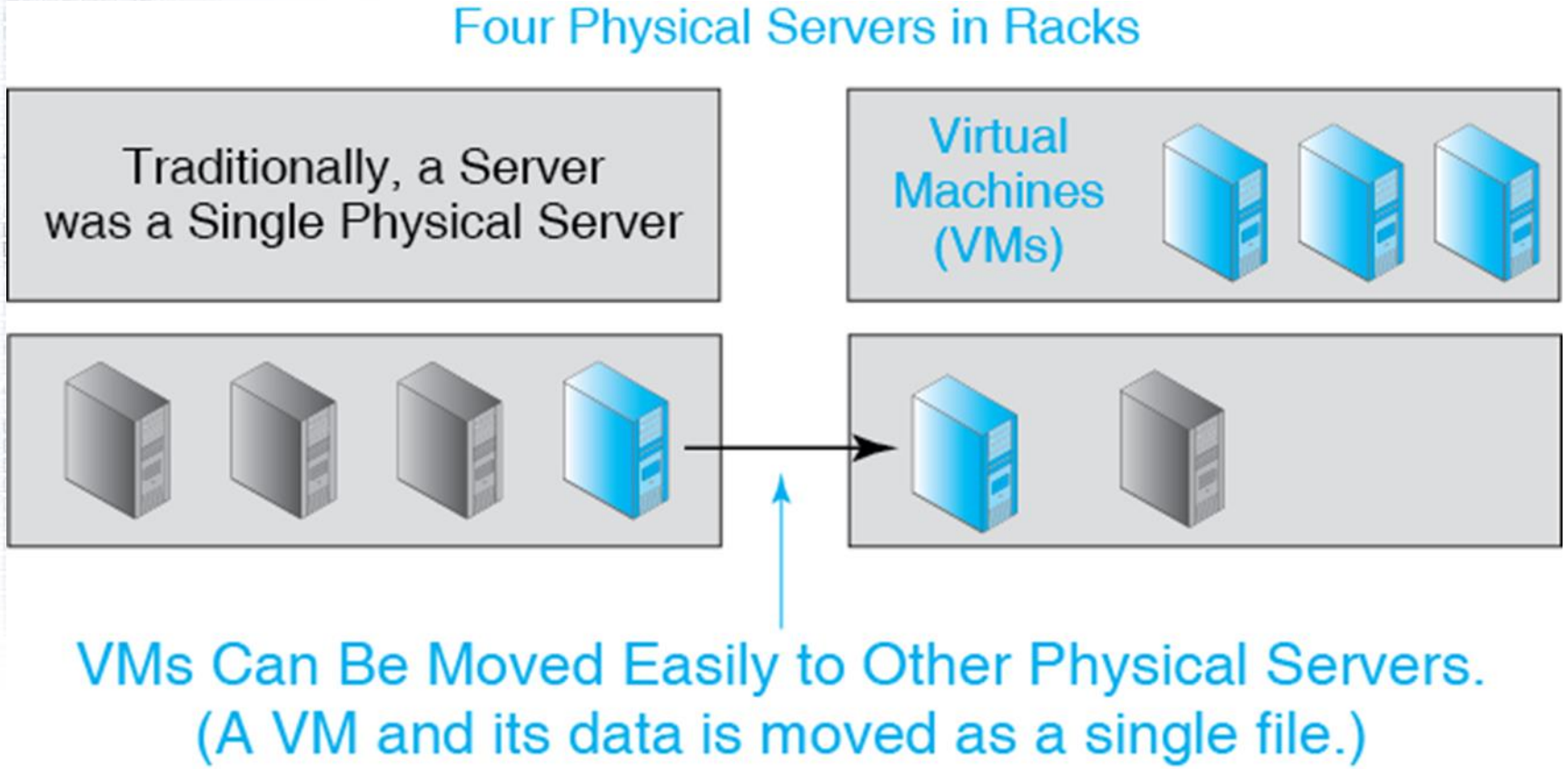




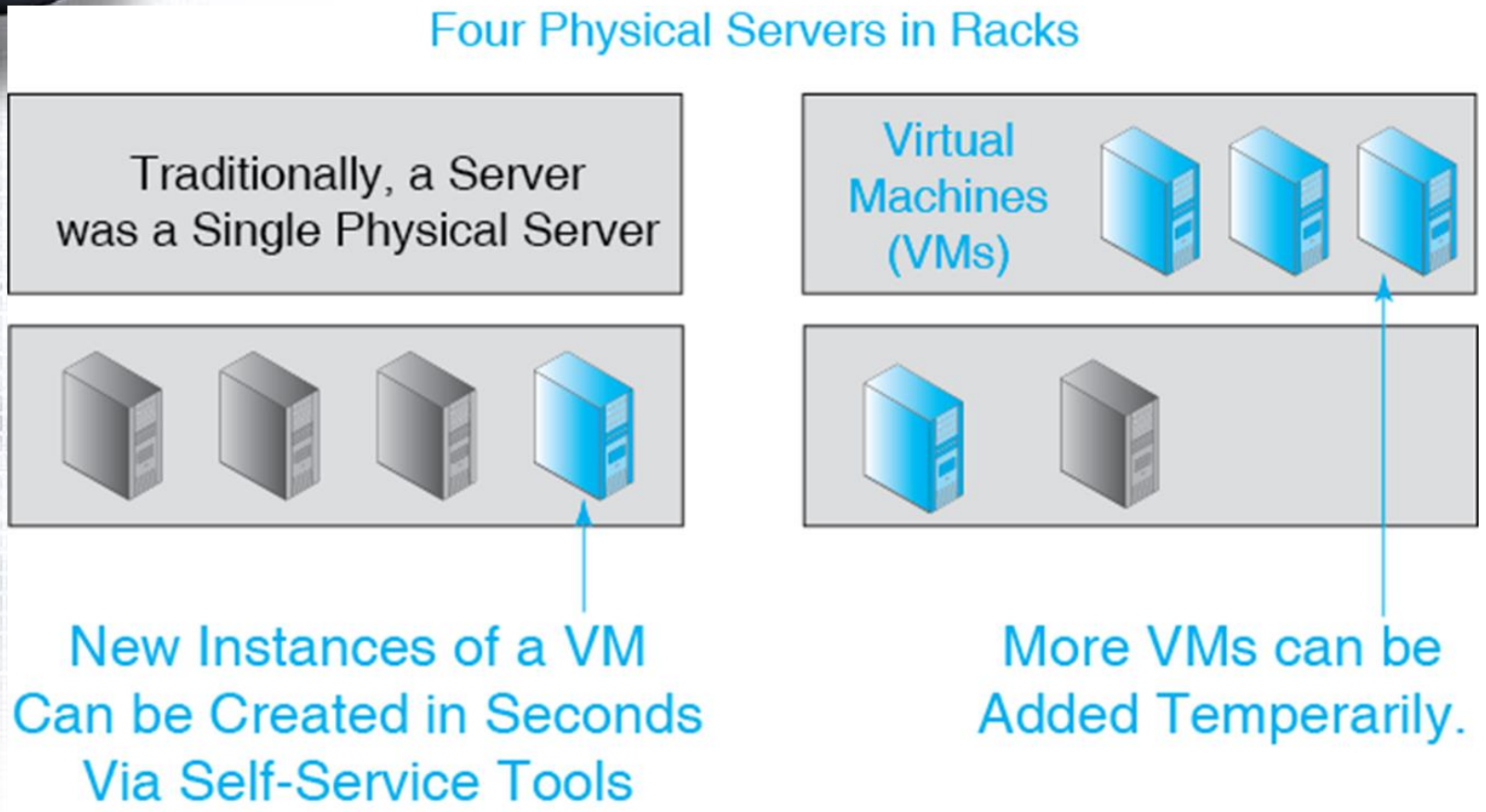
Important Cloud Technology(1/3) - Server Virtualization



Important Cloud Technology(2/3) - Server Virtualization

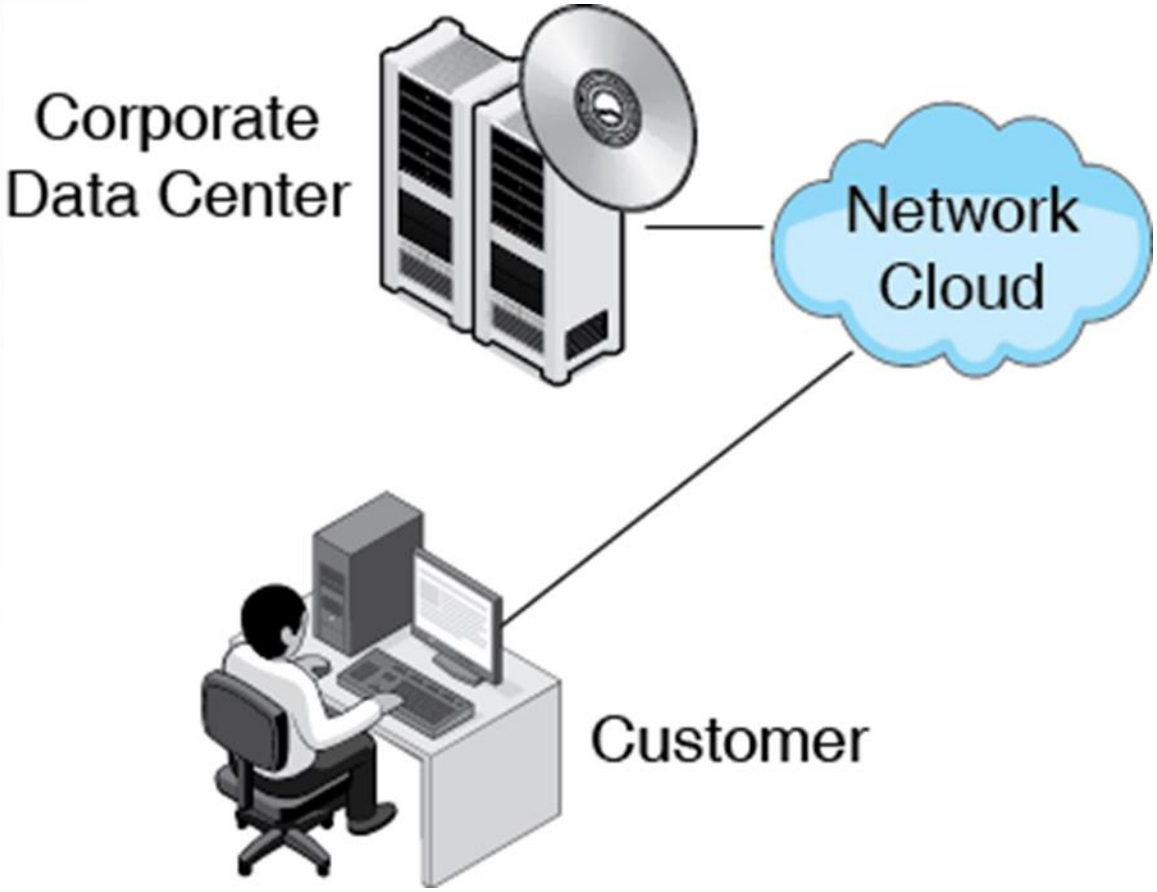


Important Cloud Technology(3/3) - Server Virtualization

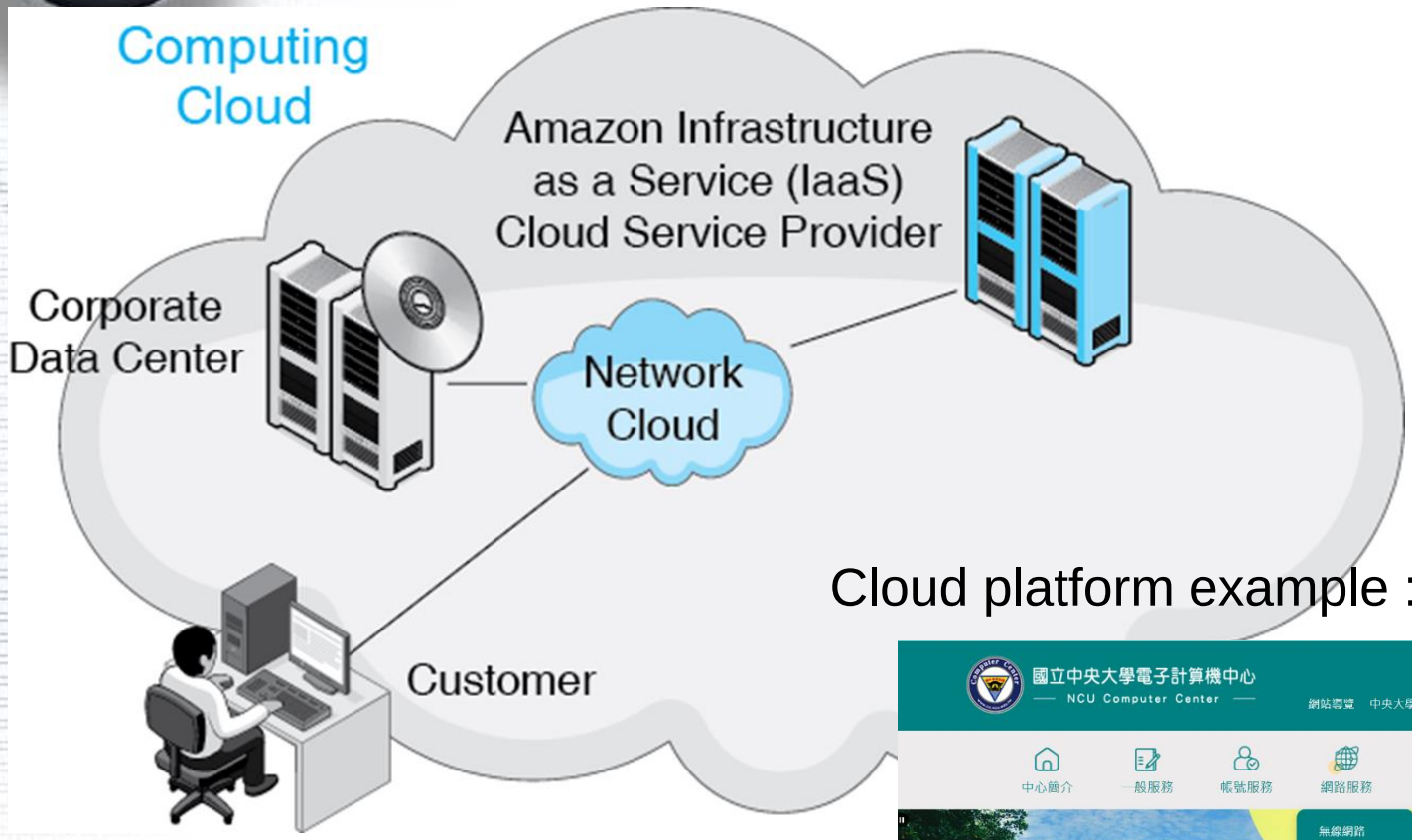


Q: 你知道 KVM、Vmware, ESXi, Vsphere, Vcenter這些名詞的關係嗎？

1.7 Traditional Internet Operations -Corporate Data Centers



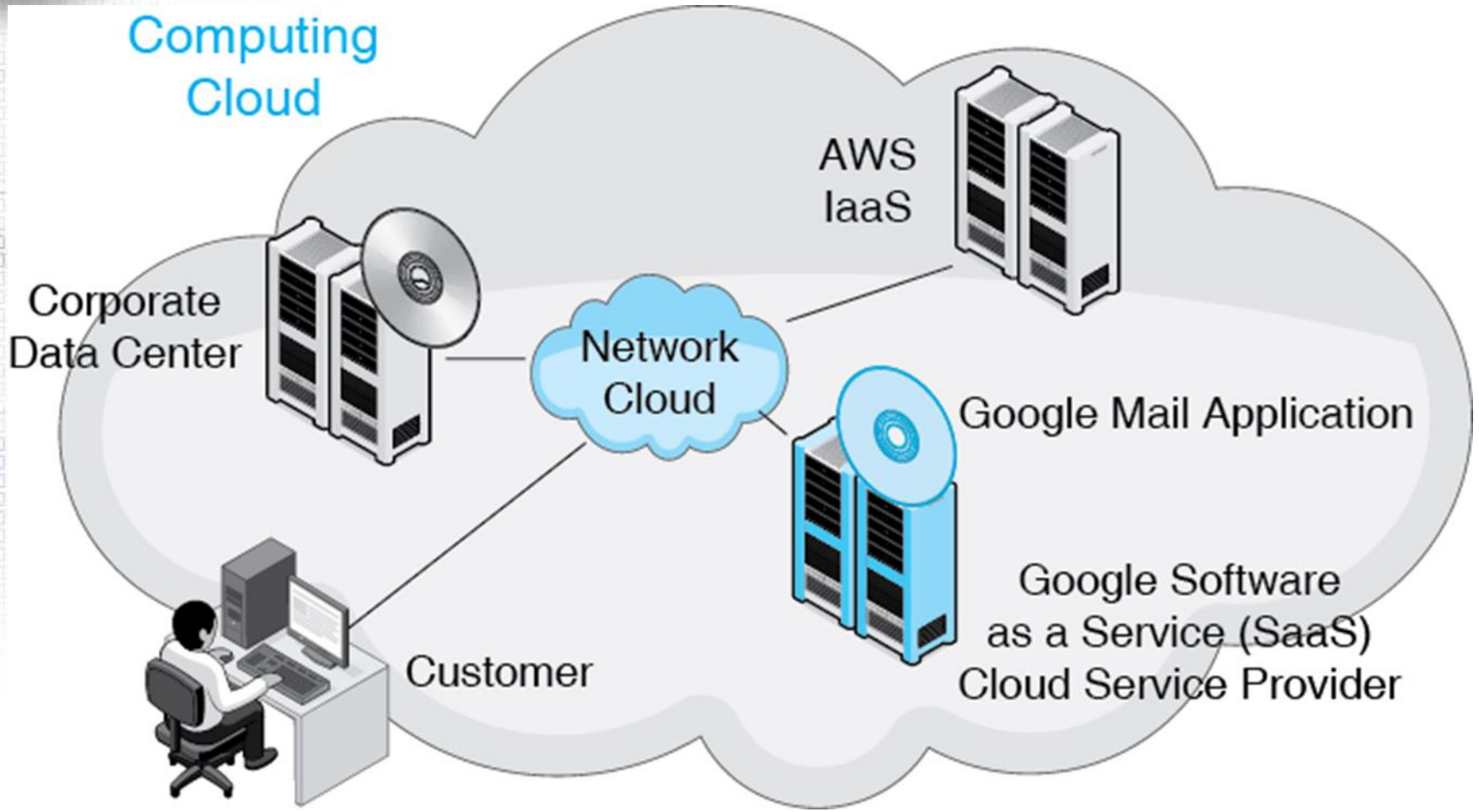
New Business in Cloud Ecosystem -Computing Cloud (IaaS)



Cloud platform example :VMware ESXi



1.7 Cloud Service Providers -IaaS vs. SaaS

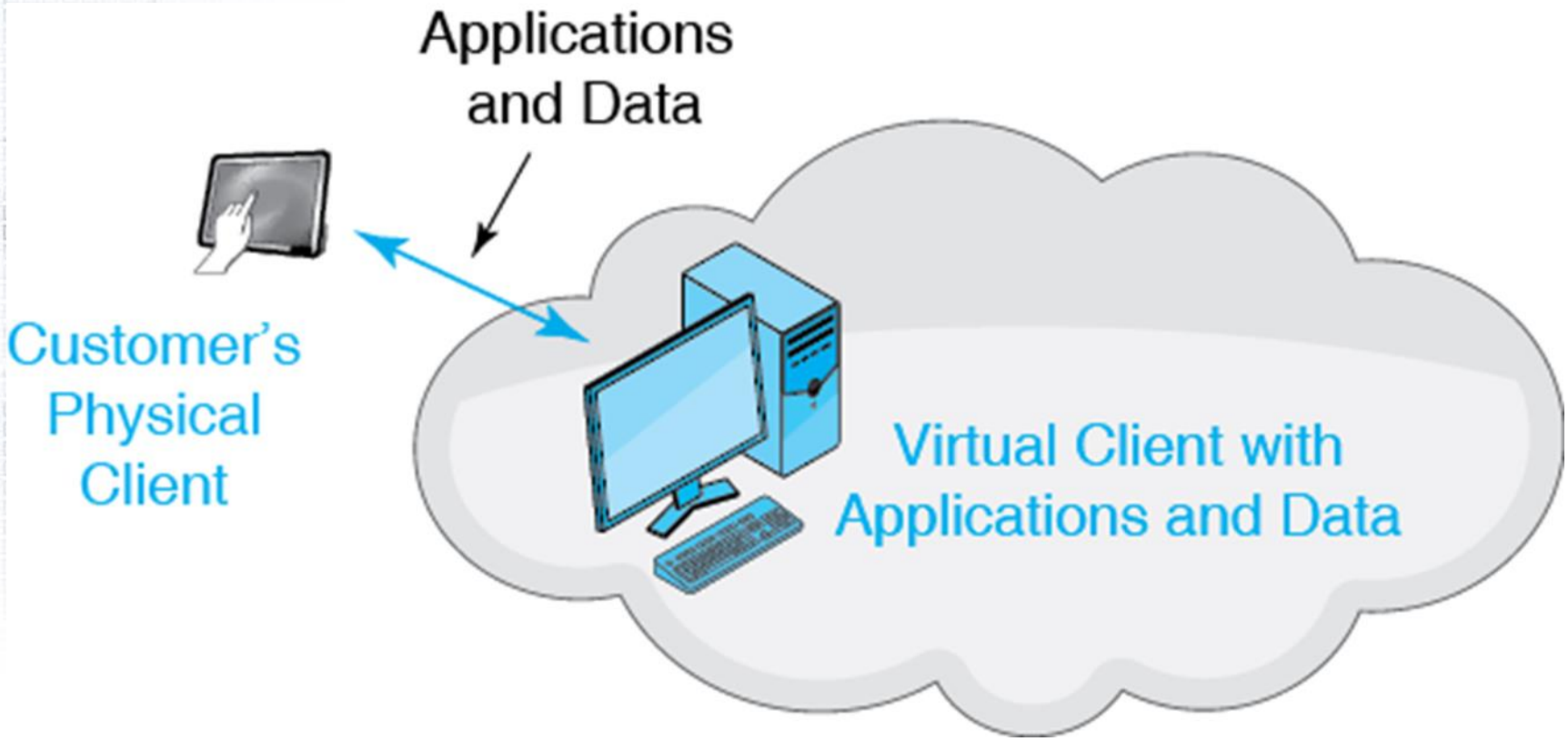


Corporate Data Centers, IaaSs, and SaaSs

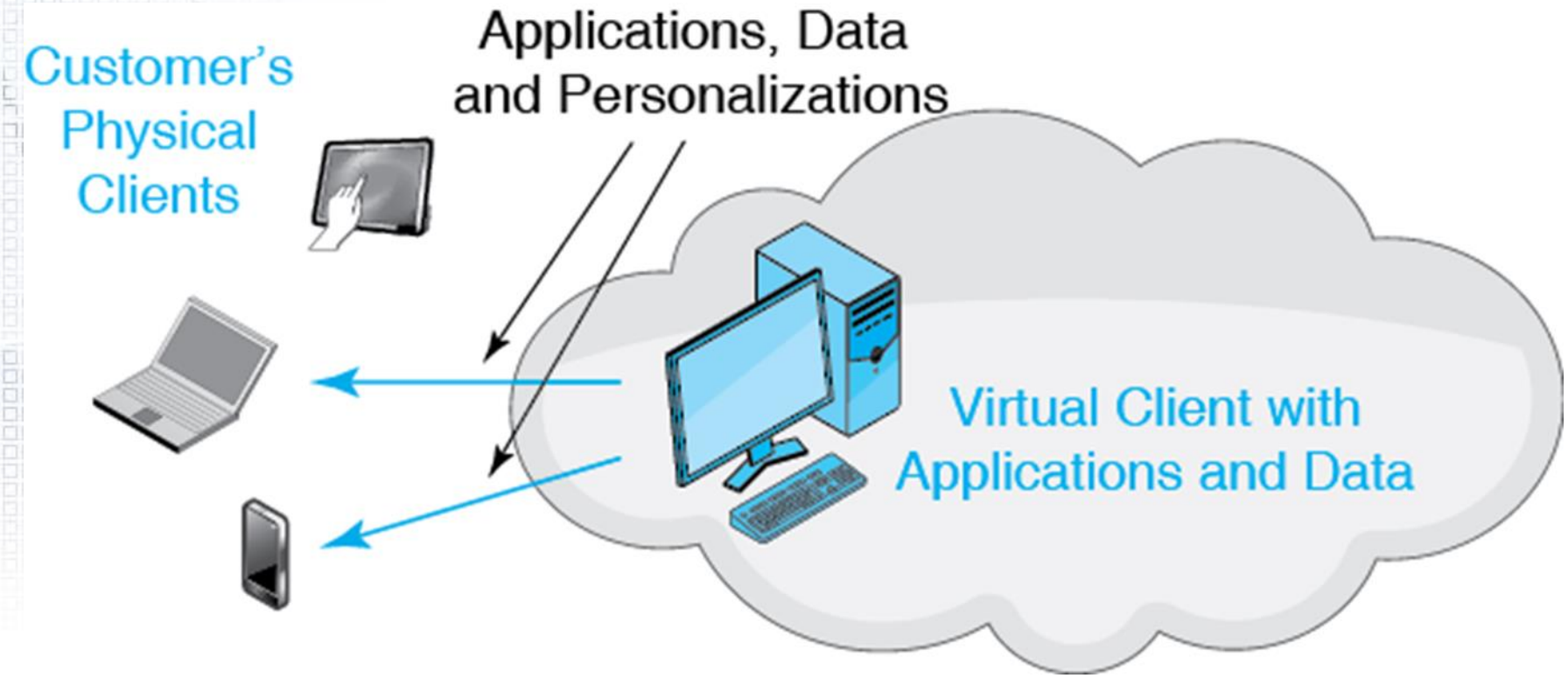
	Server Owner	Application Owner
Corporate Data Center	User (Organization)	User (Organization)
IaaS	IaaS CSP	User (Organization)
SaaS	SaaS CSP	SaaS CSP

雲端服務分類除了上述IaaS與SaaS，還有另一種稱為PaaS，你知道其意義為何嗎？

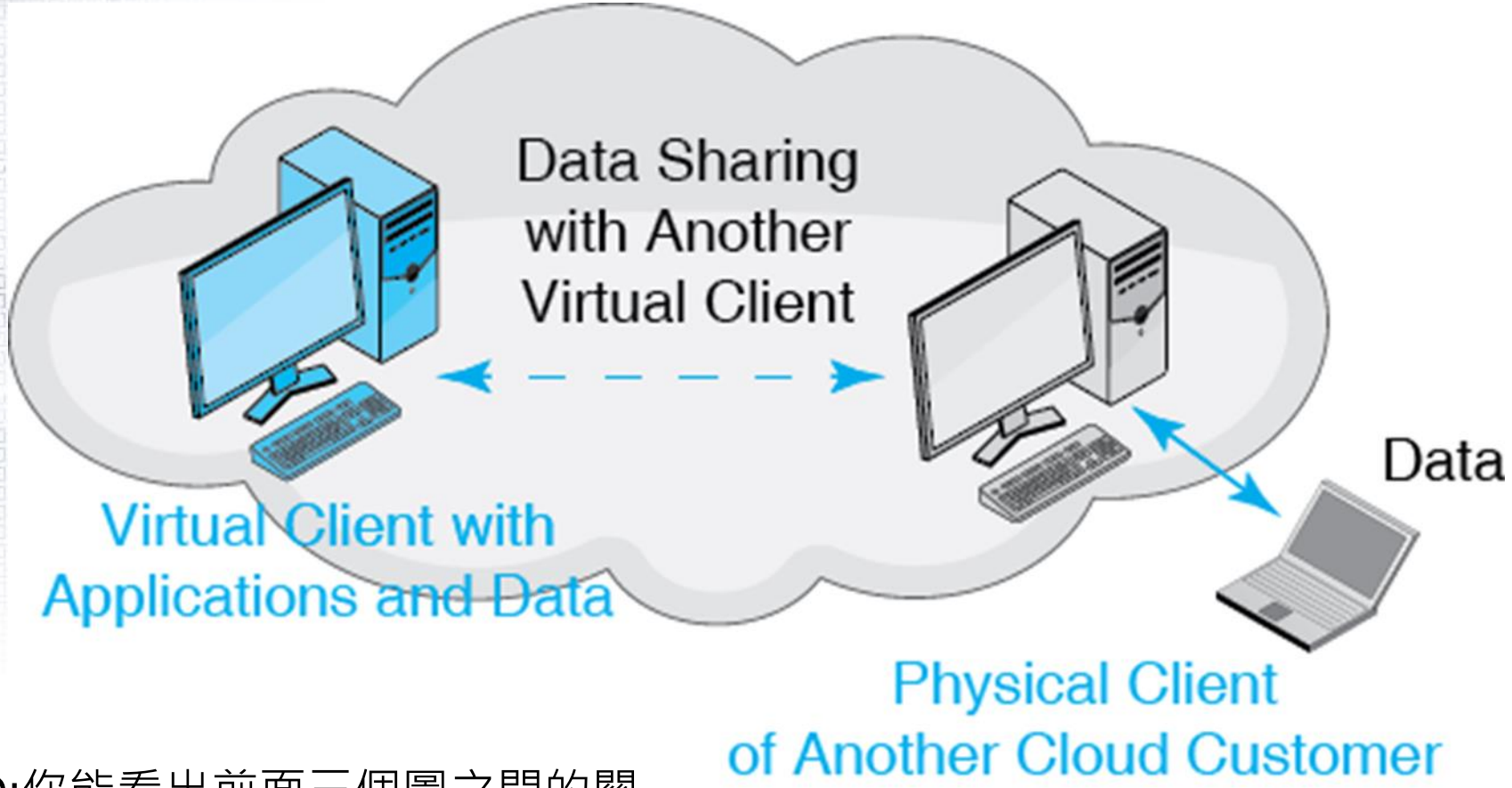
1.8 New Client-Server Relationship in Cloud Era (1/3)



1.8 New Client-Server Relationship in Cloud Era (2/3)



1.8 New Client-Server Relationship in Cloud Era (33)



Q:你能看出前面三個圖之間的關連嗎？

As a Service -Comparison

- Traditionally,
  - The organization owned and operated its own servers and application programs
  - It purchased them
  - This was Infrastructure and software as a *product*
- Cloud Services
  - No purchase for servers, applications, or both
  - Pay by amount of use, like electrical service
  - This is Infrastructure and software as a *service*

Cloud Concerns

- **Security**
  - Protection from hackers
  - The possibility that the government is reading files in the cloud
- **Stress on the Network**
  - Cloud computing drives network demand
  - This requires large increases in capacity (see next page: Application Download Times)
  - Reliability must increase

1.9 Application Download Times

Application / Speed	100 kbps	1 Mbps	10 Mbps	100 Mbs	1 Gbps
E-mail Message (250 words)	0.02 sec	—	—	—	—
Photograph (2 MB)	3 min	16 sec	2 sec	—	—
1-Hour HDTV Program (7 Mbps)	3 days	7 hr	42 min	4 min	25 sec
Backup 500 GB hard drive	16 yrs	1.9 mo	5.8 da	14 hr	2 hr

你知道TANet學術網路骨幹頻寬達到多少Gbps嗎？

(最早骨幹頻寬是1.544 Mbps，然後是44.736 Mbps)

中央資管所　陳奕明

Netflix Jumps Into the Amazon

Messages

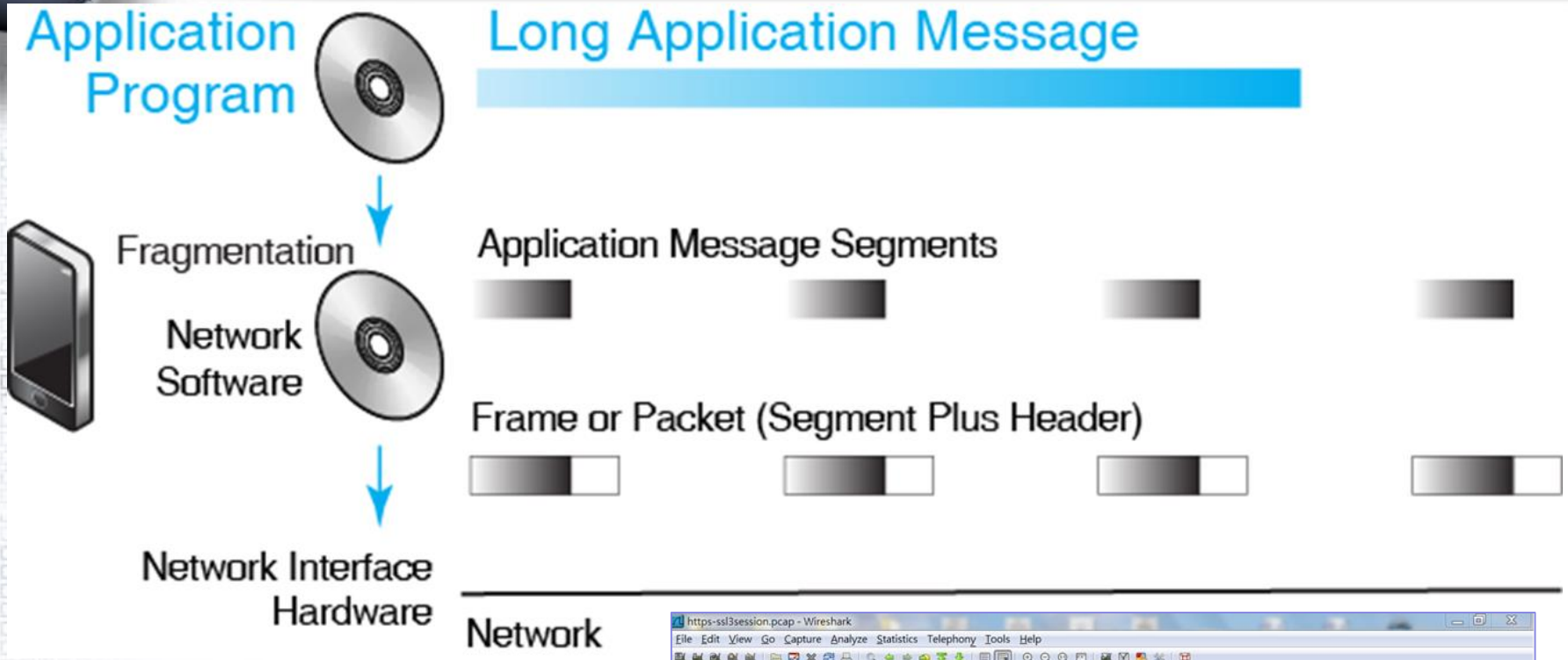
Single Networks

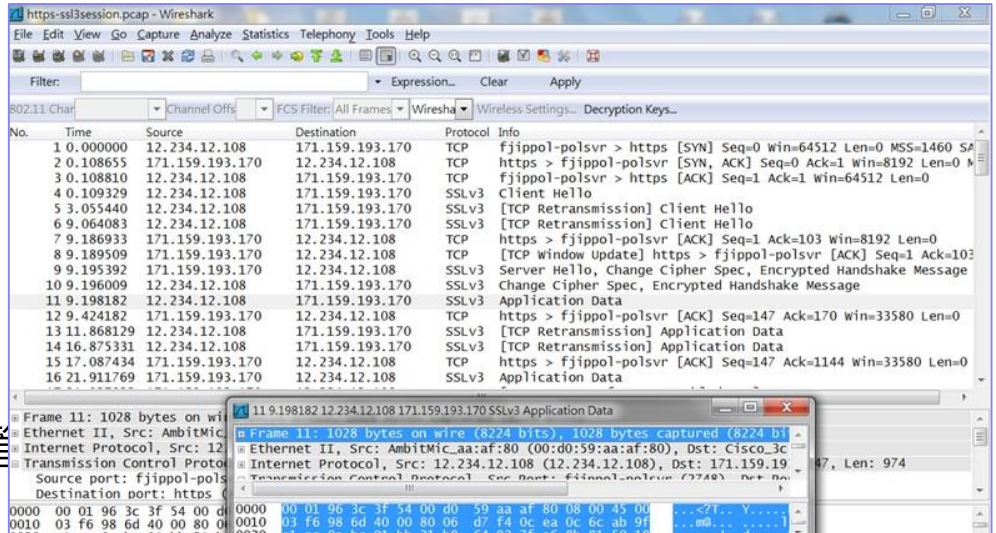
Internets

Standards Layers

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1.12 Fragments, Frames, and Packets





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Why break to fragments? (1/2)
- 1.13 Fragmentation & Error Correction

No Fragmentation



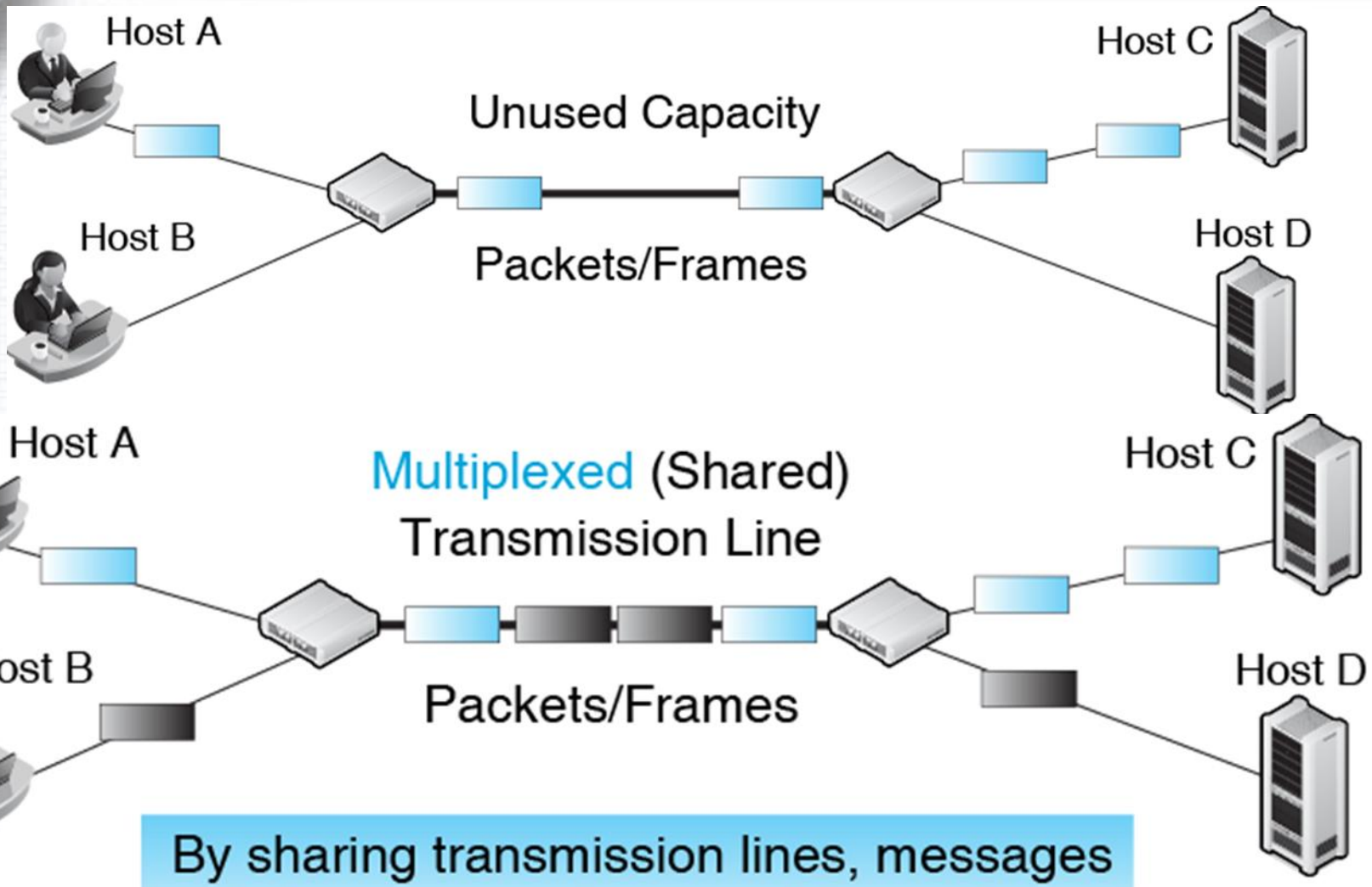
Error! Must retransmit entire message

With Fragmentation



Error! Only need to retransmit a single frame or packet

Why break to fragments? (2/2)
1.14 Fragmentation and Multiplexing



By sharing transmission lines, messages only have to pay for part of the total cost

1.15 EUI-48 Addresses (extended unique identifier)

- Used in Ethernet 802.3, Wi-Fi 802.11, and Bluetooth
  - Formerly called a media access control (MAC) address
    - Still commonly referred to as MAC addresses
  - Written as six pairs of number or letter symbols separated by dashes
    - Example A1-B2-CC-92-FF-00
  - **Not all single-network addresses follow the EUI-48 format**

Netflix Jumps Into the Amazon

Messages

Single Networks

Internets

Standards Layers

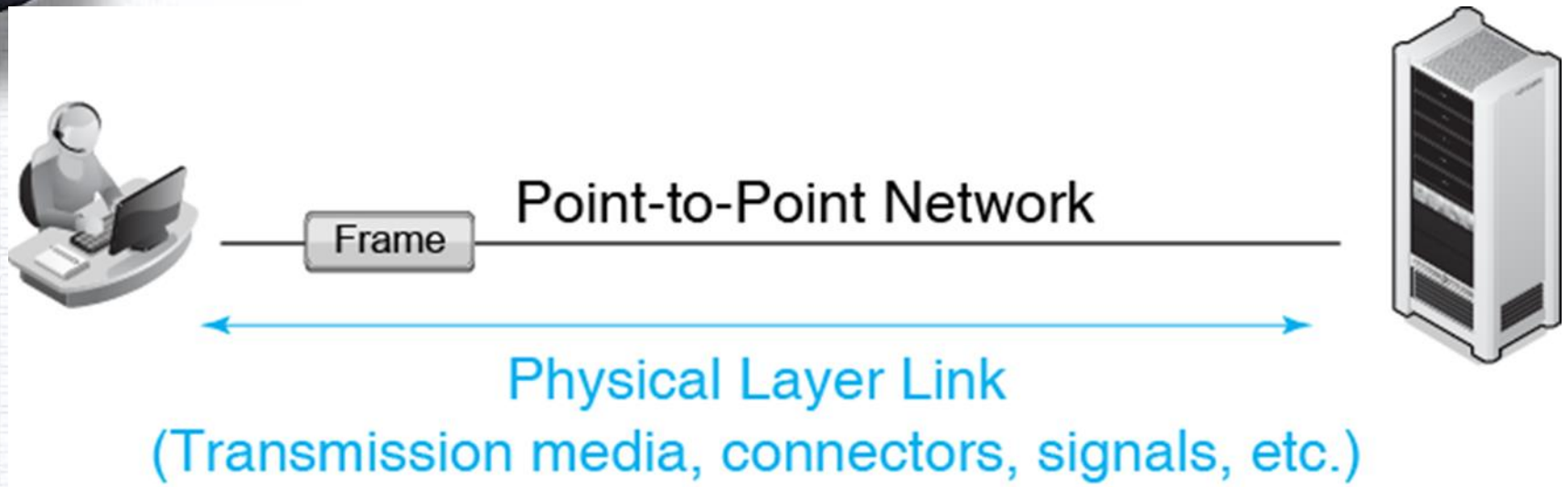
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1.16 Point-to-Point Single Network

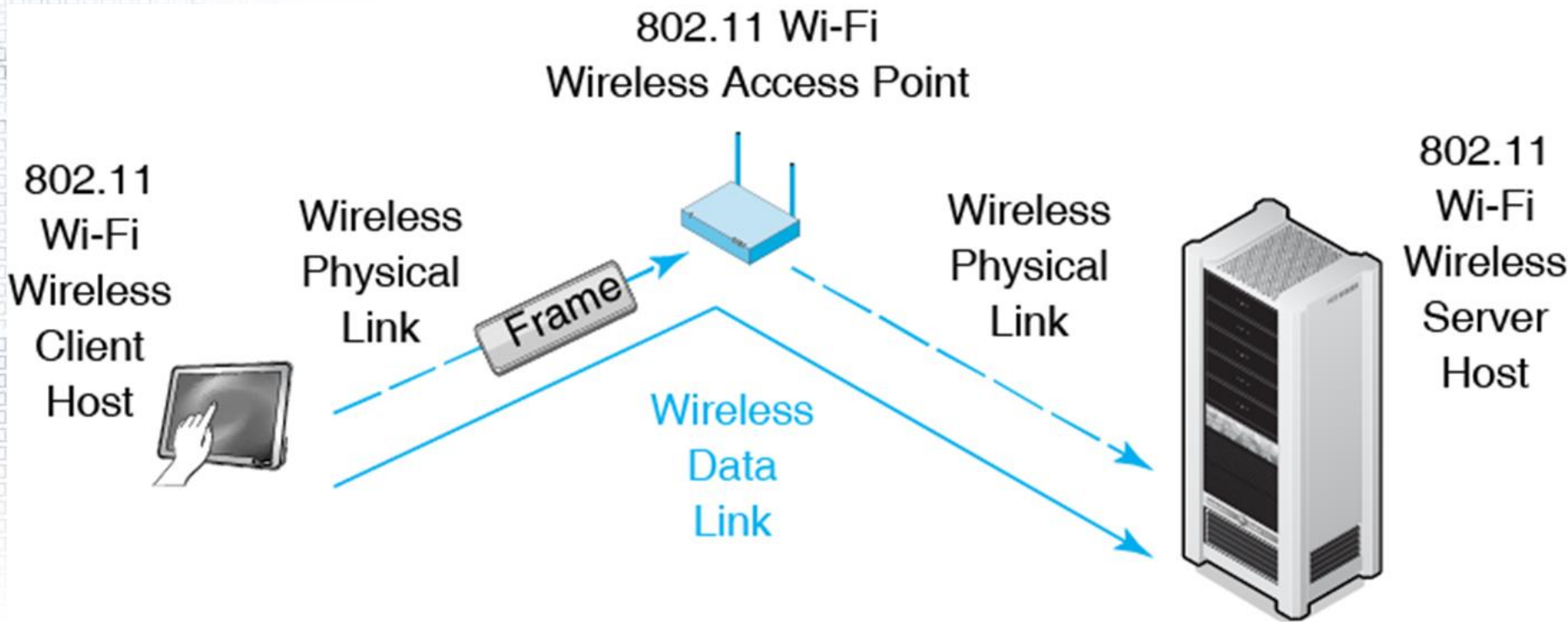


Q: Do you know how *to directly connect two NB to transfer files* without using hub or switch device?

Basic Knowledge Behind Point-to-Point Single Network

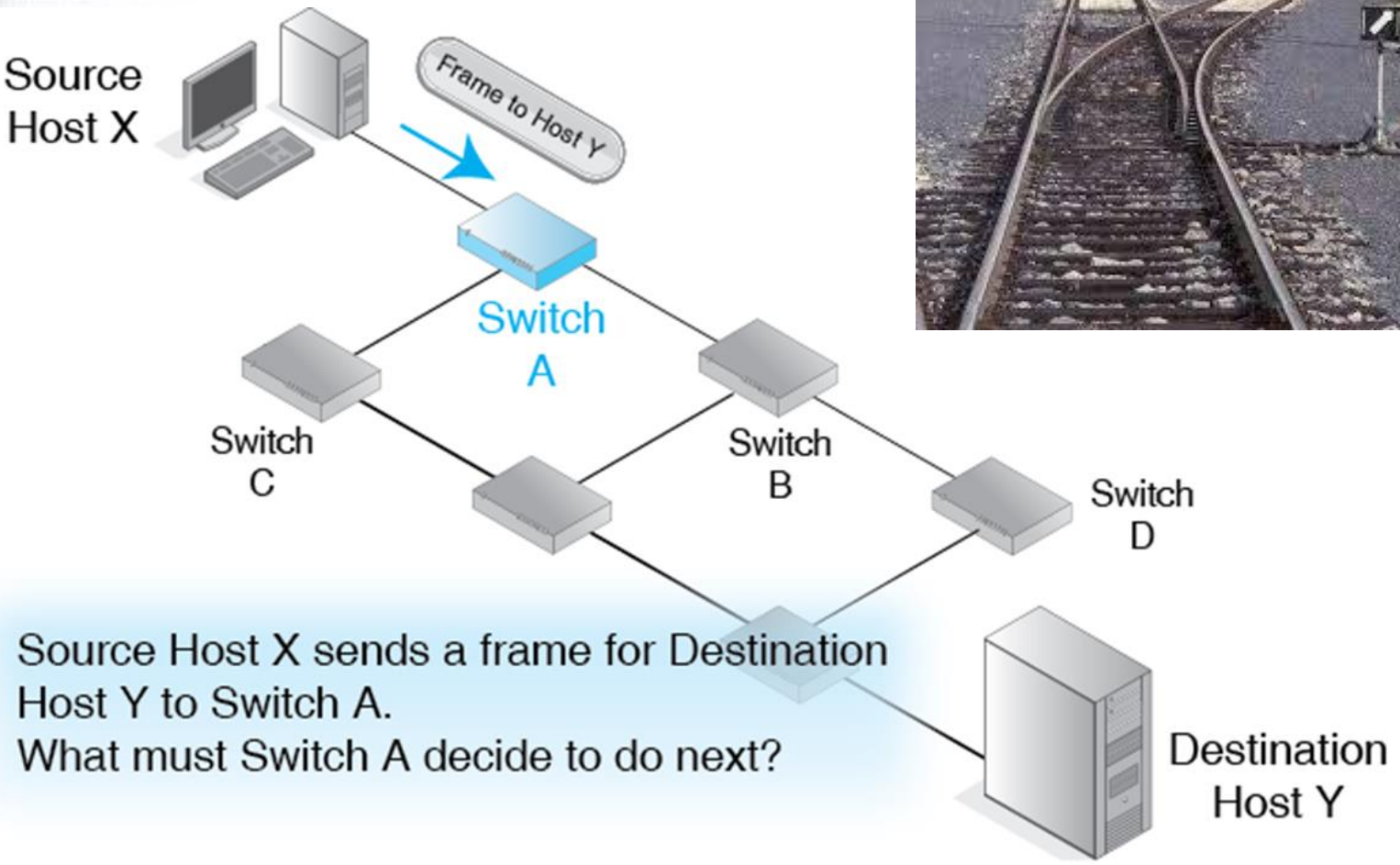


Alternative to Point-to-Point Single Network -1.18 802.11 Wireless Single Networks



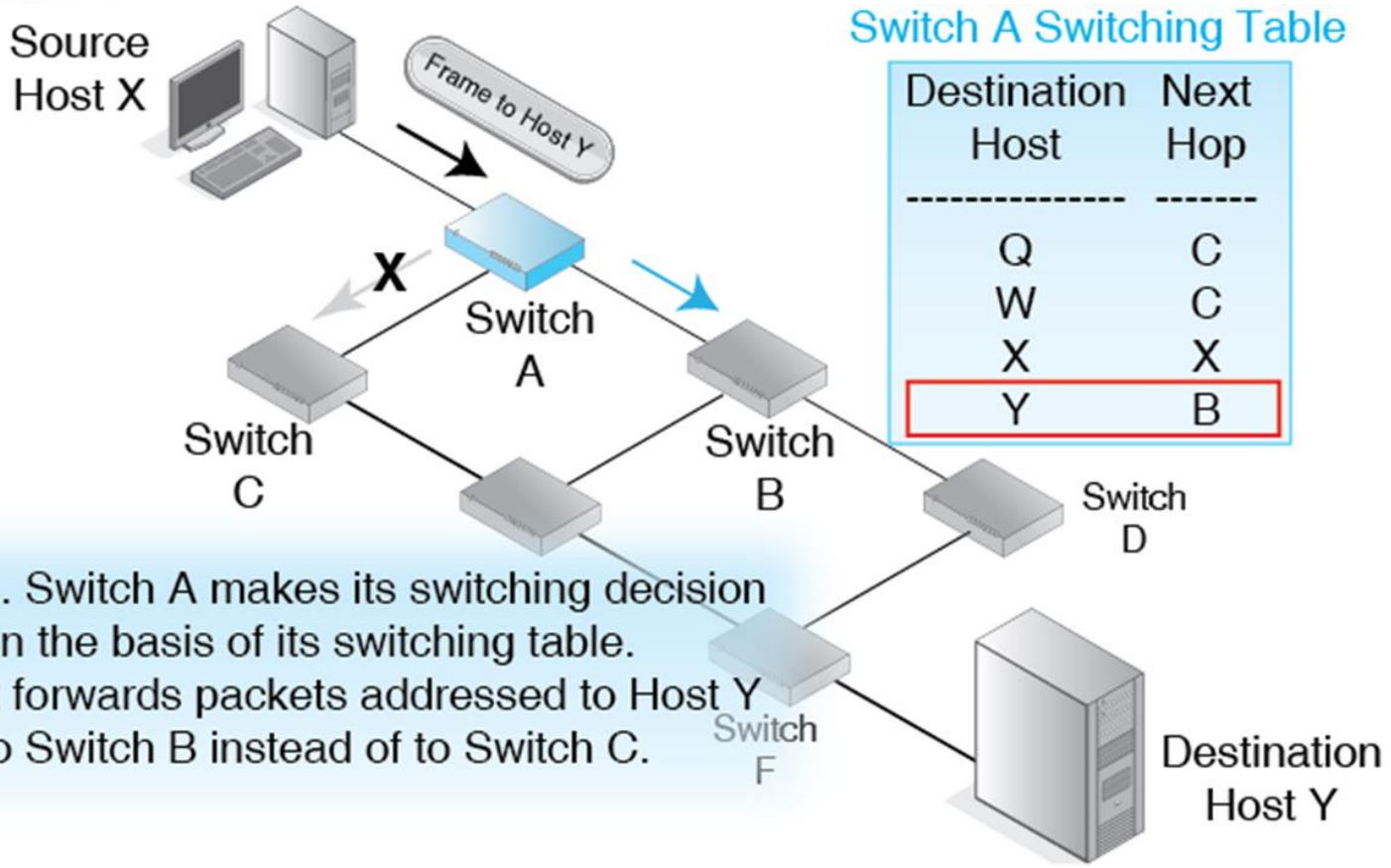
1.20 Switched Network -Why?



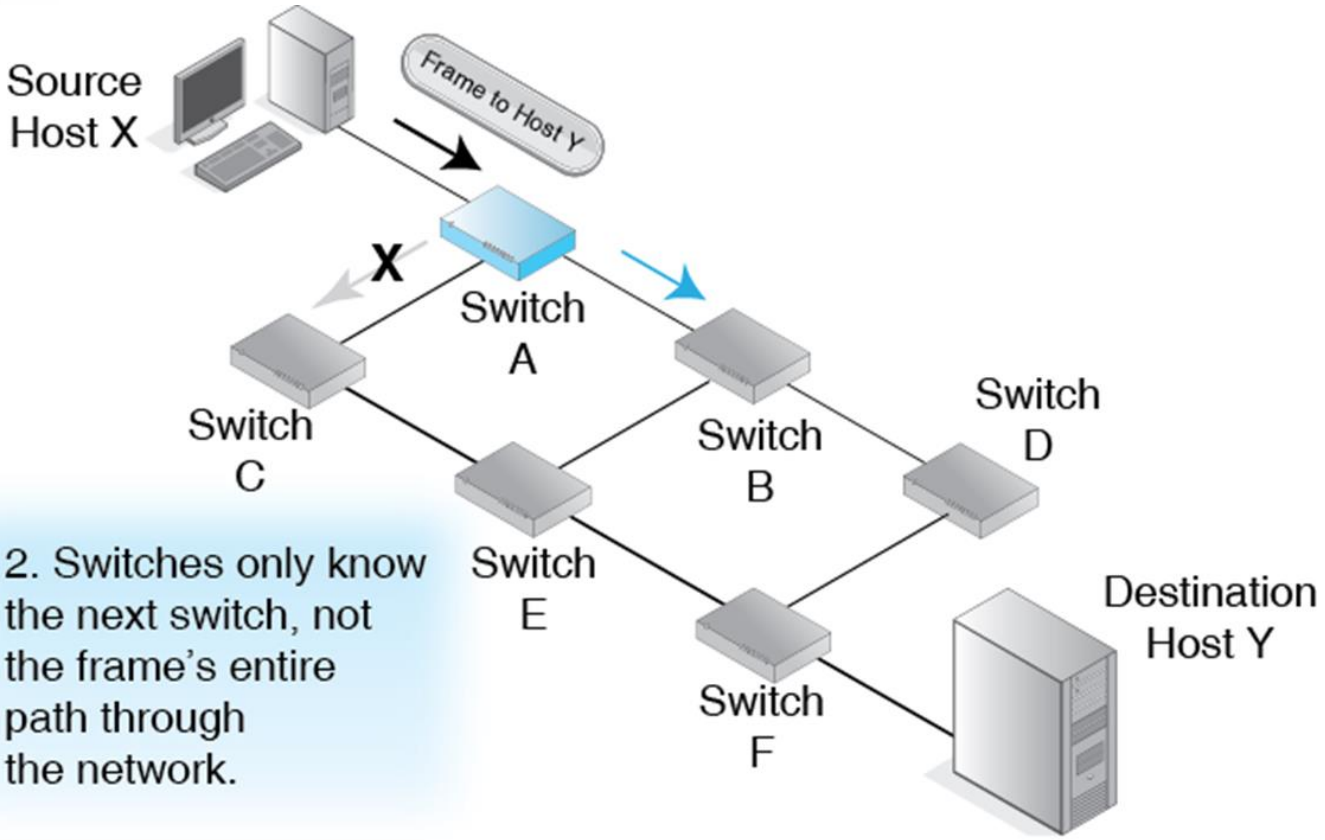


Source Host X sends a frame for Destination Host Y to Switch A.
What must Switch A decide to do next?

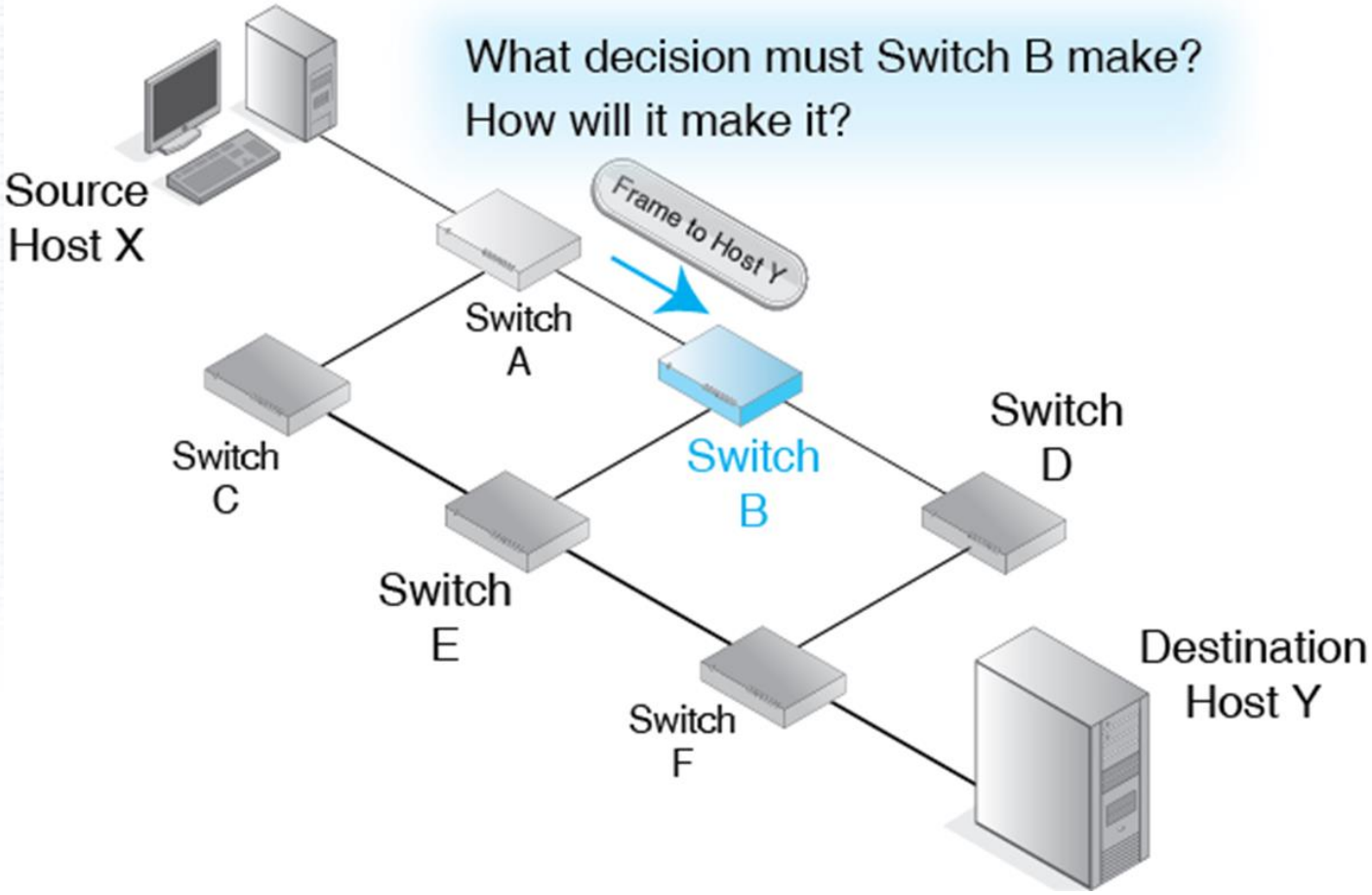
Important Concept of Switched Network: Switching Table (1/3)



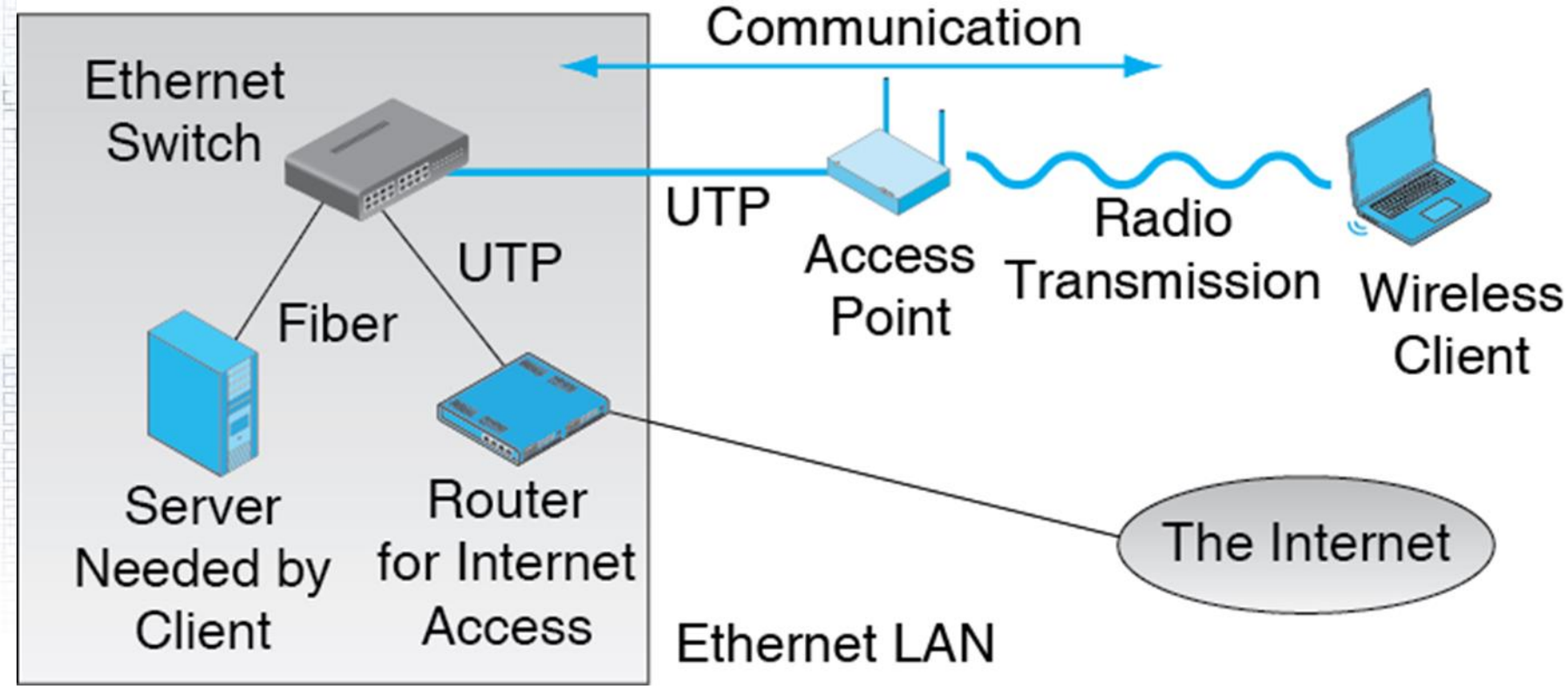
Important Concept of Switched Network: Switching Table (2/3)



Important Concept of Switched Network: Switching Table (3/3)



Common Network Connections in Today



Netflix Jumps Into the Amazon

Messages

Single Networks

Internets

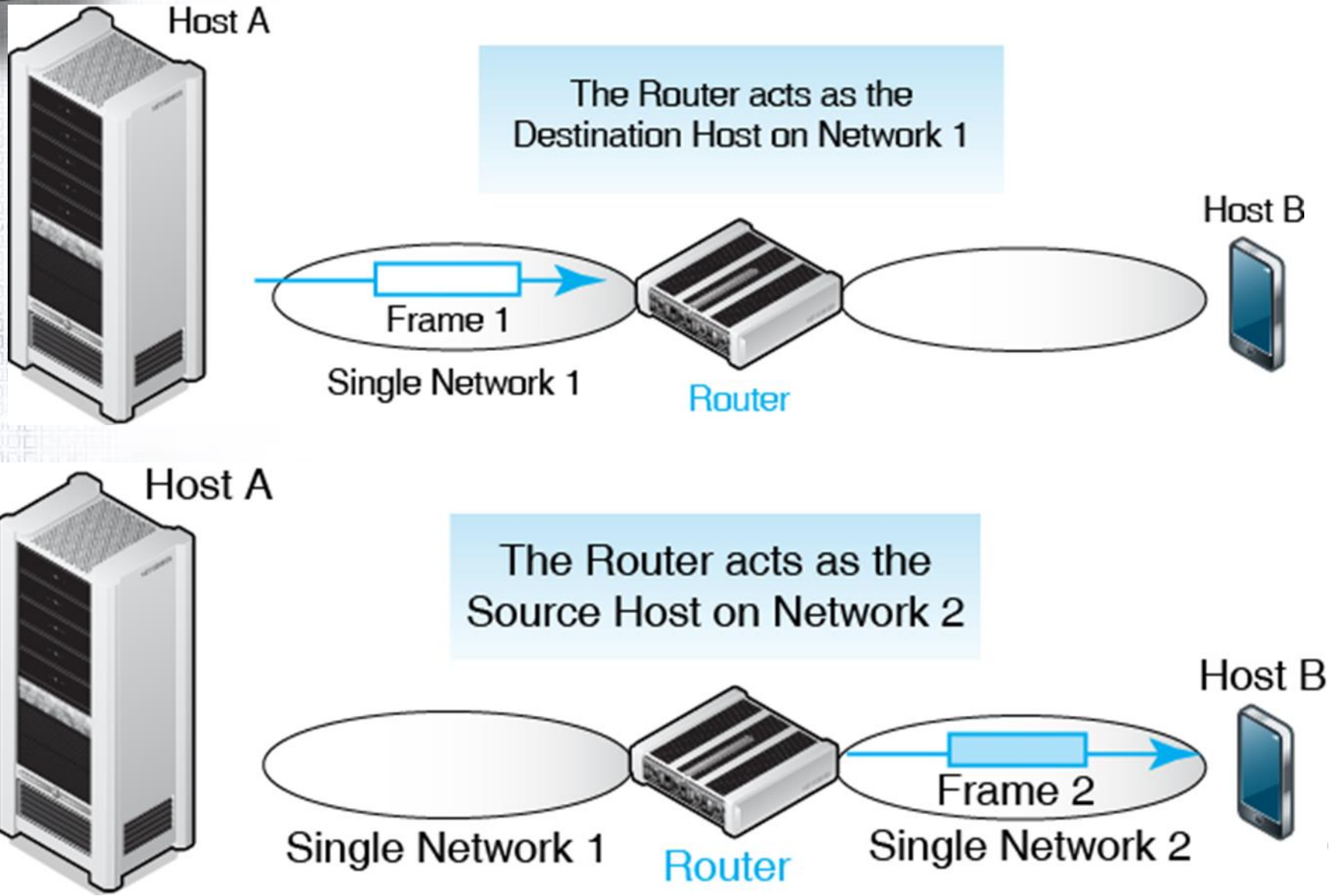
Standards Layers

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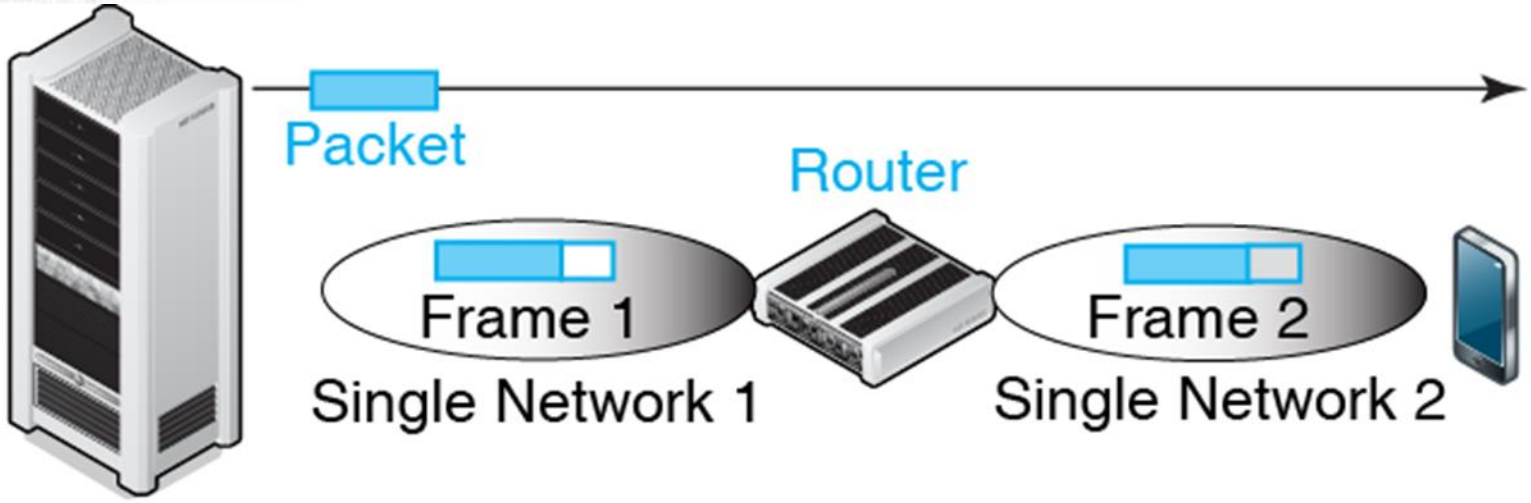
1.22 Problems Connecting Hosts on Different Single Network

- **Different Standards (例如無線網路 vs Ethernet)**
- **Overlapping Addresses**
  - Even if the two networks follow the *same* standard, a host on one network may have the same DLL (Data Link Layer) address as a host on the other network.
- **Link to Connect the Single Network**
  - If the link used the standards of one network, it would not connect to the other network for the first two reasons

1.23 Internetworking with a Router

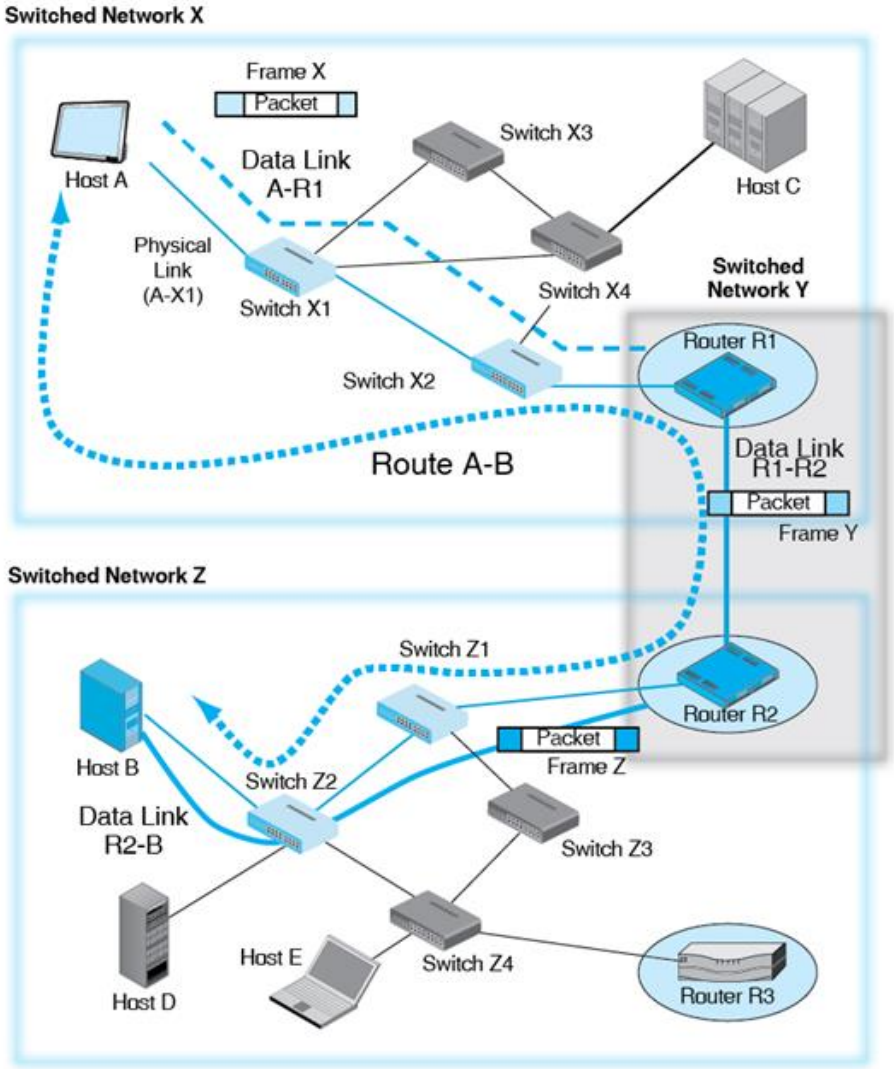


1.24 Difference Between IP Packet and Data Link Layer Frame



Packet travels all the way to the destination host.
It travels in a different frame in each single network on the way.
Frames travel as far as the next switch or host.

1.26 More Complex Internet



1.27 Repeated Concepts at the Data Link and Internet Layers

Concept	Data Link Layer	Internet Layer
Layer Number	Layer 2 (L2)	Layer 3 (L3)
Name	Data Link Layer	Internet Layer
Main Standard	Various (Ethernet 802.3, Wi-Fi 802.11, PPP, etc.)	IP
Addresses	Data Link Layer addresses (Often EUI-48 addresses)	IP addresses
Messages	Frames	Packets
Forwarding Devices	Switches, Access Points, etc.	Routers
Paths	Data Links	Routes

1.28 Common Misconception about "Packet Switching"

- Fragmentation and placing in an envelope is generically called *packet switching* ←*An important technology used in Internet*
  - However, **packets** are routed
  - And **frames** are switched
- So technically speaking, *packets are never switched*

結論：對專家而言，名詞的分辨（e.g. frame vs.packet)很重要，因為背後代表不一樣的概念。

Netflix Jumps Into the Amazon

Messages

Single Networks

Internets

Standards Layers

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1.29 General Standards Layers

Num	Name	Role
5	Application	Standardize communication between two application programs of a certain type.
4	Transport	Fragmentation and other functions.
3	Internet	Transmit a packet across an internet. Packet organization, router operation, other things needed to transmit a packet across a route in an internet.
2	Data Link	Transmit a frame across a single network. Frame organization, switch and access point operation, and other things needed to transmit a frame across a data link in a single network.
1	Physical	Transmission media, plugs and connectors, signaling.

1.30 Networking Standards Agencies and Architectures

Architecture	OSI	TCP/IP
Standard agency/agencies	ISO and ITU-T	Internet Engineering Task Force (IETF)
Architecture name	OSI	TCP/IP
Examples of standards	802.3 Ethernet, 802.11 Wi-Fi, optical fiber	TCP, IP, DNS
Layers at which dominant	Physical (1) and Data Link (2)	Internet (3) and Transport (4)

Summary

- Netflix Jumps Into the Amazon
  - Cloud Ecosystem
- Messages
  - Fragments: Frames and Packets
- Single Networks
  - Point-to-Point network vs. Switched Network
- Internets
  - The functions of Router
- Standards Layers
  - 5 Layers and OSI vs. TCP/IP

第一章分組題目

一. 2009年美國DARPA為慶祝Internet誕生40週年，特舉辦Red Balloon Challenge活動來表現40年來Internet的進展與影響力。請閱讀下列文章：***Reflecting on the DARPA Red Balloon Challenge***（以及相關論文）後，製作投影片介紹該活動並說明你的感想。此外，亦請挑出一個DARPA後來舉辦的類似挑戰賽並予以介紹（包括說明該比賽的主旨與比賽結果、後續的影響等）

二. 本章介紹Cloud Computing的原理與應用，但近年來也逐漸發展Edge computing，請上網蒐集相關文章閱讀後，製作投影片介紹Edge computing的特性、然後舉至少兩個實例說明其在產業上的應用（或者用樹莓派實做一個應用亦可）